

INDU 211: Introduction to Production & Manufacturing Systems

Problem Solution Guide 02: Linear Programming Minimization

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[!NOTE]

- **YouTube Video Tutorial:** [Watch Video 2: INDU 211 - Minimization Problem](#)
- **Video ID:** yTi70c0_cq8 · **Duration:** 19:44
- **Target Exam Scope:** ❧ **FINAL EXAM (Weeks 7–12, Chapters 14, 15, 8, 17)**
- **Curriculum Context:** Chapter 14 — Deterministic Operations Research Models

1. Problem Formulation

The linear programming problem requires finding the minimum cost point subject to an equality constraint, an inequality constraint, and non-negativity:

Minimize $Z = 4X_1 + X_2$ Subject to: $X_1 + 3X_2 = 9$ (Constraint 1: Equality) $X_1 + X_2 \leq 5$ (Constraint 2: Upper Bound) $X_1 \geq 0, X_2 \geq 0$ (

2. Graphical Representation & 1D Line-Segment Feasible Region

Step 1: Plotting Constraint Lines

1. Constraint 1 ($X_1 + 3X_2 = 9$):

- If $X_1 = 0 \implies 3X_2 = 9 \implies X_2 = 3 \implies (0, 3)$
- If $X_2 = 0 \implies X_1 = 9 \implies (9, 0)$
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Critical Mathematical Note

: Because this is a strict equality ($=$), feasible solutions must lie **strictly ON** this line segment, not over a 2D surface area.

2. Constraint 2 ($X_1 + X_2 \leq 5$):

- If $X_1 = 0 \implies X_2 = 5 \implies (0, 5)$
- If $X_2 = 0 \implies X_1 = 5 \implies (5, 0)$
- Feasible solutions must lie on or below this line in the direction of the origin $(0, 0)$.

3. Non-Negativity Constraints ($X_1 \geq 0, X_2 \geq 0$):

- Restricts solutions to the first quadrant (X_1 -axis and X_2 -axis).

Step 2: Determining the Feasible Region Extremities (Corner Points)

Because of the equality constraint, the feasible space collapses into a **1-dimensional line segment** along $X_1 + 3X_2 = 9$:

- **Extreme Point A (Left Endpoint):** At the intersection with the X_2 -axis ($X_1 = 0$):

$$0 + 3X_2 = 9 \implies X_2 = 3 \implies \text{Point A} = (0, 3)$$

Verify against Constraint 2: $0 + 3 = 3 \leq 5$ (Valid).

- **Extreme Point B (Right Endpoint):** At the intersection of the two constraint boundary lines:

$$\begin{cases} X_1 + 3X_2 = 9 \\ X_1 + X_2 = 5 \end{cases} \implies X_1 = 5 - X_2$$

Substitute X_1 into Constraint 1:

$$(5 - X_2) + 3X_2 = 9 \implies 5 + 2X_2 = 9 \implies 2X_2 = 4 \implies X_2 = 2$$

$$X_1 = 5 - 2 = 3 \implies \text{Point B} = (3, 2)$$

Verify against Constraint 1: $3 + 3(2) = 9$ (Valid).

3. Objective Function Propagation & Optimal Solution

Iso-Cost Line Propagation:

Set the objective function to an arbitrary value, say $Z = 5$:

$$4X_1 + X_2 = 5 \implies \text{Intercepts: } (0, 5) \text{ and } (1.25, 0)$$

- The slope of the objective line is $m = -4$.
- For a **minimization** problem, propagate the iso-cost line from the upper right inward toward the origin $(0, 0)$.
- The point on the feasible line segment that the iso-cost line touches at the lowest Z value is the global optimum.

Corner Point Evaluation Table:

Corner Point	Coordinates (X_1, X_2)	Objective Value $Z = 4X_1 + X_2$	Status
Point A	$(0, 3)$	$Z = 4(0) + 3 = \mathbf{3}$	GLOBAL MINIMUM (OPTIMAL)
Point B	$(3, 2)$	$Z = 4(3) + 2 = \mathbf{14}$	Sub-optimal

Optimal Solution:

$$X_1^* = 0, \quad X_2^* = 3, \quad Z^* = 3$$

4. Part 3.2.3: Characterization of All Feasible Solutions

Question:

Identify all feasible solutions of this linear programming problem using a mathematical expression or concise explanation.

- **Mathematical Definition:** $S = \{ (X_1, X_2) \in \mathbb{R}^2; X_1 + 3X_2 = 9, 0 \leq X_1 \leq 3, 0 \leq X_2 \leq 3 \}$
- **Concise Explanation:**
"The feasible region of this linear program is not a two-dimensional polygon, but rather a one-dimensional straight line segment lying along $X_1 + 3X_2 = 9$ bounded between the extreme endpoints $(0, 3)$ and $(3, 2)$."